

## EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at [source.efymag.com](http://source.efymag.com)

### PARTS LIST

#### Semiconductors:

|          |                              |
|----------|------------------------------|
| IC1      | - 7805, 5V regulator         |
| IC2      | - PIC 16F73 microcontroller  |
| IC3, IC4 | - H11D1 optoisolator         |
| D1, D2   | - 1N4007 rectifier diode     |
| D3, D4   | - BA159 fast switching diode |
| ZD1, ZD2 | - 5V, 0.5W zener diode       |
| T1, T2   | - MJE13005 power transistor  |

Resistors (all 1/4-watt,  $\pm 5\%$  carbon, unless stated otherwise):

|        |                                                                    |
|--------|--------------------------------------------------------------------|
| R1, R2 | - 100-ohm                                                          |
| R3, R4 | - 1-kilo-ohm                                                       |
| R5, R6 | - 10-ohm, 5W                                                       |
| R7-R9  | - 10-kilo-ohm                                                      |
| R10    | - 680-ohm                                                          |
| VR1    | - 10-kilo-ohm, 16mm plastic shaft potentiometer with on/off switch |

#### Capacitors:

|        |                                  |
|--------|----------------------------------|
| C1     | - 470 $\mu$ F, 25V electrolytic  |
| C2     | - 0.1 $\mu$ F ceramic            |
| C3, C4 | - 15pF ceramic                   |
| C5, C6 | - 470 $\mu$ F, 400V electrolytic |
| C7, C8 | - 1nF, 1kV ceramic               |

#### Miscellaneous:

|            |                                 |
|------------|---------------------------------|
| F1         | - 1.5A glass fuse               |
| BATT.1     | - 9V PP3 battery with connector |
| CON1, CON2 | - 2-pin connector terminal      |
| S1         | - On/off switch (VR1)           |
| XTAL1      | - 20MHz crystal oscillator      |
|            | - Fan                           |
|            | - Heat-sinks for IC1, T1 and T2 |
|            | - 28-pin socket for PIC IC      |
|            | - 6-pin socket for IC3 and IC4  |

ages. You need to use an isolation transformer for the scope, ensuring that you don't touch the oscilloscope's input pin. Otherwise, a small transformer (230V primary to 9V secondary) can be connected in lieu of or in parallel to the fan and the secondary voltage (9V) observed for waveshape and frequency. This method was used by the author for creating the positive and negative DC rails and it showed a good sine wave shape with some floating ripples of 50Hz, as the mains supply is not full-wave rectified and the

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capacitors used are also only 470 $\mu$ F electrolytic. This does not affect the fan's working speed anyhow.

### Construction and testing

An actual-size PCB layout for the fan speed increasing regulator is shown in Fig. 7 and its components layout in Fig. 8. After assembling the circuit, enclose it in a suitable box.

Follow the testing steps given below:

1. After checking the circuit for proper connections, connect a 9V battery and switch on the circuit using VR1. LED1 on the board should glow.

2. Check pins 12 and 13 of PIC16F73 chip for pulse output. For this, use a logic probe or a CRO, if available. Pulse-widths vary as you vary VR1. You can notice a small voltage of 0.2V or so on pin 1 of each of H11D1 chip with respect to the negative of the 9V battery supply. That means the variable PWM sine waveform is correctly generated by the chip.

3. Now connect a fan to CON2 connector. Make sure you turn on the inbuilt fan switch.

4. Connect 230V mains supply to CON1. It is advisable to use a variable auto transformer and apply 140V first, to see the fan running. Varying VR1 clockwise will increase the fan's speed in steps. The fan can be switched off by turning back VR1 and switching off its built-in on/off switch.

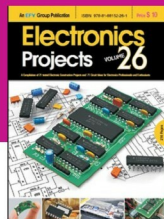
5. If the fan does not run, the fuse blows up or 10-ohm resistors heat up. Check the transistors using a multimeter for emitter-collector short-circuit on both sides in that case.

**Caution.** Since this circuit works on 230V AC mains voltage, extreme caution is required while testing it. Wearing surgical rubber gloves is advisable to avoid electric shock. **EFY**

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